2

Compiled Data

Timing

Clock Time (ns)

	LFSR 4	LSFR 8	LSFR 12
14nm	0.1	0.1	0.1
32nm	0.24	0.3	0.3
45nm	0.23	0.3	0.3

We can see that the design itself has little influence on the clock speed, although more complicated means slightly slower times. A smaller process node appears to correlate with higher clock speeds.

Power

Cell Leakage Power (nW)

	LFSR 4	LSFR 8	LSFR 12
14nm	0.8155532	1.9496	2.5555
32nm	5121.1	12039.4	17383.9
45nm	635.2694	1316.8	1954.6

Total Dynamic Power (uW)

	LFSR 4	LSFR 8	LSFR 12
14nm	2.7321	6.3137	7.2258
32nm	156.4370	274.2990	386.0458
45nm	619.6028	990.9634	1461.8

It appears that the larger the design, the more power leakage there will be. However, the 14nm node appears to have the least leakage, with 45nm next, and the 32nm last. This is likely due to the fact that these are different manufacturers with varying levels of refinement on their process node.

For total dynamic power however, it appears that size scales linearly with power, and the smaller process nodes exponentially utilize less power than the larger ones.

Area

Total Cell Area

	LFSR 4	LSFR 8	LSFR 12
14nm	3.241200	6.171600	8.302800
32nm	52.353664	103.944897	148.928385
45nm	80.719600	165.662899	240.750899

And for cell area, it is faily obvious that smaller nodes are smaller and smaller designs are smaller. The process node and size do not appear to be inversely linearly related however, as the 14nm node seems significantly smaller than the other sizes.

LFSR 4

14nm

Timing

```
Point                          Trans      Incr       Path
-----------------------------------------------------------
clock clk (rise edge)                     0.00       0.00
clock network delay (ideal)               0.00       0.00
data_reg_0_/clocked_on (**FFGEN**)
                                0.00      0.00       0.00 r
data_reg_0_/Q (**FFGEN**)       0.00      0.00       0.00 r
data[0] (out)                   0.00      0.00       0.00 r
data arrival time                                    0.00

clock clk (rise edge)                     0.10       0.10
clock network delay (ideal)               0.00       0.10
output external delay                    -0.10       0.00
data required time                                   0.00
-----------------------------------------------------------
data required time                                   0.00
data arrival time                                    0.00
-----------------------------------------------------------
slack (MET)                                          0.00
```

Power

```
  Cell Internal Power  =   1.1025 uW   (40%)
  Net Switching Power  =   1.6296 uW   (60%)
                         ---------
Total Dynamic Power    =   2.7321 uW  (100%)

Cell Leakage Power     = 815.5532 pW


                 Internal         Switching           Leakage
Total
Power Group      Power            Power               Power
Power   (   %     )  Attrs
------------------------------------------------------------------------
--------------------------
io_pad             0.0000            0.0000            0.0000
0.0000  (   0.00%)
memory             0.0000            0.0000            0.0000
0.0000  (   0.00%)
black_box          0.0000            0.0000            0.0000
0.0000  (   0.00%)
clock_network      0.0000            0.0000            0.0000
0.0000  (   0.00%)  i
register           0.0000            1.2216            0.0000
1.2216  (  44.70%)
sequential         0.0000            0.0000            0.0000
0.0000  (   0.00%)
combinational      1.1025            0.4081          815.5532
1.5113  (  55.30%)
------------------------------------------------------------------------
--------------------------
Total              1.1025 uW         1.6296 uW       815.5532 pW
2.7329 uW
```

Area

```
Number of ports:                            10
Number of nets:                             21
Number of cells:                            15
Number of combinational cells:              11
Number of sequential cells:                  4
```

```
Number of macros/black boxes:                0
Number of buf/inv:                           1
Number of references:                        6

Combinational area:                   3.241200
Buf/Inv area:                         0.177600
Noncombinational area:                0.000000
Macro/Black Box area:                 0.000000
Net Interconnect area:      undefined  (No wire load specified)

Total cell area:                      3.241200
Total area:                 undefined
```

32nm

Timing

```
  Point                              Trans       Incr        Path
  ------------------------------------------------------------------
  clock clk (rise edge)                          0.00        0.00
  clock network delay (ideal)                    0.00        0.00
  data_reg_2_/CLK (DFFASRX1_RVT)
                                      0.00       0.00        0.00 r
  data_reg_2_/QN (DFFASRX1_RVT)
                                      0.03       0.08        0.08 r
  U16/Y (INVX0_RVT)                   0.02       0.03        0.11 f
  U17/SO (HADDX1_RVT)                 0.02       0.08        0.19 r
  data_reg_0_/D (DFFASRX1_RVT)
                                      0.02       0.01        0.21 r
  data arrival time                                          0.21

  clock clk (rise edge)                          0.24        0.24
  clock network delay (ideal)                    0.00        0.24
  data_reg_0_/CLK (DFFASRX1_RVT)                 0.00        0.24 r
  library setup time                            -0.03        0.21
  data required time                                         0.21
  ------------------------------------------------------------------
  data required time                                         0.21
  data arrival time                                         -0.21
  ------------------------------------------------------------------
  slack (MET)                                                0.00
```

Power

```
  Cell Internal Power  = 147.1922 uW   (94%)
  Net Switching Power  =   9.2448 uW    (6%)
                         ---------
Total Dynamic Power    = 156.4370 uW  (100%)

Cell Leakage Power     =   5.1211 uW


                 Internal         Switching           Leakage
Total
Power Group      Power            Power               Power
Power   (   %    )  Attrs
--------------------------------------------------------------------------

--------------------------
io_pad             0.0000            0.0000            0.0000
0.0000  (   0.00%)
memory             0.0000            0.0000            0.0000
0.0000  (   0.00%)
black_box          0.0000            0.0000            0.0000
0.0000  (   0.00%)
clock_network    126.2969            0.0000            0.0000
126.2969  (  78.17%)  i
register          13.8878            4.0886        4.0659e+06
22.0423  (  13.64%)
sequential         0.0000            0.0000            0.0000
0.0000  (   0.00%)
combinational      7.0075            5.1562        1.0552e+06
13.2189  (   8.18%)
--------------------------------------------------------------------------

--------------------------
Total            147.1922 uW         9.2448 uW     5.1211e+06 pW
161.5581 uW
```

Area

```
Number of ports:                           10
Number of nets:                            25
Number of cells:                           18
Number of combinational cells:             14
Number of sequential cells:                 4
```

```
Number of macros/black boxes:                0
Number of buf/inv:                           5
Number of references:                        4

Combinational area:                  21.856384
Buf/Inv area:                         6.353600
Noncombinational area:               30.497280
Macro/Black Box area:                 0.000000
Net Interconnect area:                4.082460

Total cell area:                     52.353664
Total area:                          56.436124
```

45nm

Timing

```
  Point                             Trans       Incr        Path
  ------------------------------------------------------------------
  clock clk (rise edge)                         0.00        0.00
  clock network delay (ideal)                   0.00        0.00
  data_reg_2_/CLK (DFFSR)            0.00       0.00        0.00 r
  data_reg_2_/Q (DFFSR)              0.03       0.12        0.12 r
  U19/Y (XNOR2X1)                    0.02       0.03        0.15 f
  U18/Y (INVX1)                      0.00       0.00        0.16 r
  data_reg_0_/D (DFFSR)              0.00       0.00        0.16 r
  data arrival time                                         0.16

  clock clk (rise edge)                         0.23        0.23
  clock network delay (ideal)                   0.00        0.23
  data_reg_0_/CLK (DFFSR)                       0.00        0.23 r
  library setup time                           -0.07        0.16
  data required time                                        0.16
  ------------------------------------------------------------------
  data required time                                        0.16
  data arrival time                                        -0.16
  ------------------------------------------------------------------
  slack (MET)                                               0.00
```

Power

```
  Cell Internal Power  = 584.9819 uW   (94%)
  Net Switching Power  =  34.6209 uW    (6%)
                         ---------
Total Dynamic Power    = 619.6028 uW  (100%)

Cell Leakage Power     = 635.2694 nW


                 Internal         Switching           Leakage
Total
Power Group      Power            Power               Power
Power   (   %    )  Attrs
--------------------------------------------------------------------------
--------------------------
io_pad             0.0000            0.0000            0.0000
0.0000  (   0.00%)
memory             0.0000            0.0000            0.0000
0.0000  (   0.00%)
black_box          0.0000            0.0000            0.0000
0.0000  (   0.00%)
clock_network      0.4075            0.0000            0.0000
0.4075  (  65.71%)  i
register           0.1397        1.1494e-02          434.4520
0.1516  (  24.44%)
sequential         0.0000            0.0000            0.0000
0.0000  (   0.00%)
combinational  3.7783e-02        2.3127e-02          200.8174
6.1111e-02  (   9.85%)
--------------------------------------------------------------------------
--------------------------
Total              0.5850 mW     3.4621e-02 mW       635.2694 nW
0.6202 mW
```

Area

```
Number of ports:                           10
Number of nets:                            29
Number of cells:                           23
Number of combinational cells:             19
Number of sequential cells:                 4
Number of macros/black boxes:               0
Number of buf/inv:                         10
```

```
Number of references:                    6

Combinational area:                39.421199
Buf/Inv area:                      15.956200
Noncombinational area:             41.298401
Macro/Black Box area:               0.000000
Net Interconnect area:      undefined  (No wire load specified)

Total cell area:                   80.719600
Total area:                 undefined
```

LFSR 8

14nm

Timing

```
  Point                          Trans      Incr       Path
  -----------------------------------------------------------
  clock clk (rise edge)                     0.00       0.00
  clock network delay (ideal)               0.00       0.00
  data_reg_0_/clocked_on (**FFGEN**)
                                  0.00      0.00       0.00 r
  data_reg_0_/Q (**FFGEN**)       0.00      0.00       0.00 r
  data[0] (out)                   0.00      0.00       0.00 r
  data arrival time                                    0.00

  clock clk (rise edge)                     0.10       0.10
  clock network delay (ideal)               0.00       0.10
  output external delay                    -0.10       0.00
  data required time                                   0.00
  -----------------------------------------------------------
  data required time                                   0.00
  data arrival time                                    0.00
  -----------------------------------------------------------
  slack (MET)                                          0.00
```

Power

```
  Cell Internal Power  =   3.6580 uW   (58%)
  Net Switching Power  =   2.6557 uW   (42%)
                         ---------
```

```
Total Dynamic Power    =   6.3137 uW  (100%)

Cell Leakage Power     =   1.9496 nW


                 Internal         Switching           Leakage
Total
Power Group      Power            Power               Power
Power   (   %    )  Attrs
--------------------------------------------------------------------------------
--------------------------
io_pad             0.0000            0.0000             0.0000
0.0000  (   0.00%)
memory             0.0000            0.0000             0.0000
0.0000  (   0.00%)
black_box          0.0000            0.0000             0.0000
0.0000  (   0.00%)
clock_network      0.0000            0.0000             0.0000
0.0000  (   0.00%)  i
register           0.0000            1.8396             0.0000
1.8396  (  29.13%)
sequential         0.0000            0.0000             0.0000
0.0000  (   0.00%)
combinational      3.6580            0.8161         1.9496e+03
4.4761  (  70.87%)
--------------------------------------------------------------------------------
--------------------------
Total              3.6580 uW         2.6557 uW      1.9496e+03 pW
6.3157 uW
```

Area

```
Number of ports:                           18
Number of nets:                            37
Number of cells:                           27
Number of combinational cells:             19
Number of sequential cells:                 8
Number of macros/black boxes:               0
Number of buf/inv:                          1
Number of references:                       6

Combinational area:                  6.171600
Buf/Inv area:                        0.177600
```

```
Noncombinational area:              0.000000
Macro/Black Box area:               0.000000
Net Interconnect area:     undefined  (No wire load specified)

Total cell area:                    6.171600
Total area:                undefined
```

32nm

Timing

```
  Point                             Trans     Incr      Path
  -----------------------------------------------------------
  clock clk (rise edge)                       0.00      0.00
  clock network delay (ideal)                 0.00      0.00
  data_reg_7_/CLK (DFFASRX1_RVT)
                                     0.00     0.00      0.00 r
  data_reg_7_/Q (DFFASRX1_RVT)
                                     0.02     0.10      0.10 r
  U33/SO (HADDX1_RVT)                0.02     0.08      0.19 f
  U34/Y (XOR3X2_RVT)                 0.03     0.06      0.25 r
  data_reg_0_/D (DFFASRX1_RVT)
                                     0.03     0.01      0.26 r
  data arrival time                                     0.26

  clock clk (rise edge)                       0.30      0.30
  clock network delay (ideal)                 0.00      0.30
  data_reg_0_/CLK (DFFASRX1_RVT)              0.00      0.30 r
  library setup time                         -0.04      0.26
  data required time                                    0.26
  -----------------------------------------------------------
  data required time                                    0.26
  data arrival time                                    -0.26
  -----------------------------------------------------------
  slack (MET)                                           0.00
```

Power

```
  Cell Internal Power  = 257.8664 uW   (94%)
  Net Switching Power  =  16.4326 uW    (6%)
                         ---------
Total Dynamic Power    = 274.2990 uW  (100%)
```

```
Cell Leakage Power     =  12.0394 uW


                  Internal         Switching           Leakage
Total
Power Group       Power            Power               Power
Power   (   %    )  Attrs
--------------------------------------------------------------------------
--------------------------
io_pad              0.0000           0.0000              0.0000
0.0000  (   0.00%)
memory              0.0000           0.0000              0.0000
0.0000  (   0.00%)
black_box           0.0000           0.0000              0.0000
0.0000  (   0.00%)
clock_network     202.0316           0.0000              0.0000
202.0316  (  70.56%)  i
register           24.0165           7.9207          8.1347e+06
40.0718  (  13.99%)
sequential          0.0000           0.0000              0.0000
0.0000  (   0.00%)
combinational      31.8183           8.5119          3.9048e+06
44.2349  (  15.45%)
--------------------------------------------------------------------------
--------------------------
Total             257.8664 uW       16.4326 uW     1.2039e+07 pW
286.3384 uW
```

Area

```
Number of ports:                              18
Number of nets:                               41
Number of cells:                              29
Number of combinational cells:                21
Number of sequential cells:                    8
Number of macros/black boxes:                  0
Number of buf/inv:                             3
Number of references:                          6

Combinational area:                    42.950336
Buf/Inv area:                           3.812160
Noncombinational area:                 60.994560
```

```
Macro/Black Box area:                 0.000000
Net Interconnect area:                7.911347

Total cell area:                    103.944897
Total area:                         111.856244
```

45nm

Timing

```
  Point                          Trans       Incr        Path
  -----------------------------------------------------------
  clock clk (rise edge)                      0.00        0.00
  clock network delay (ideal)                0.00        0.00
  data_reg_1_/CLK (DFFSR)         0.00       0.00        0.00 r
  data_reg_1_/Q (DFFSR)           0.04       0.13        0.13 r
  U31/Y (XNOR2X1)                 0.05       0.06        0.19 r
  U30/Y (XNOR2X1)                 0.02       0.04        0.22 f
  U54/Y (INVX1)                   0.00       0.00        0.23 r
  data_reg_0_/D (DFFSR)           0.00       0.00        0.23 r
  data arrival time                                      0.23

  clock clk (rise edge)                      0.30        0.30
  clock network delay (ideal)                0.00        0.30
  data_reg_0_/CLK (DFFSR)                    0.00        0.30 r
  library setup time                        -0.07        0.23
  data required time                                     0.23
  -----------------------------------------------------------
  data required time                                     0.23
  data arrival time                                     -0.23
  -----------------------------------------------------------
  slack (MET)                                            0.00
```

Power

```
  Cell Internal Power  = 925.7891 uW   (93%)
  Net Switching Power  =  65.1743 uW    (7%)
                         ---------
Total Dynamic Power    = 990.9634 uW  (100%)

Cell Leakage Power     =   1.3168 uW
```

```
                 Internal         Switching           Leakage
Total
Power Group      Power            Power               Power
Power   (   %    )  Attrs
--------------------------------------------------------------------------
--------------------------
io_pad             0.0000            0.0000            0.0000
0.0000  (   0.00%)
memory             0.0000            0.0000            0.0000
0.0000  (   0.00%)
black_box          0.0000            0.0000            0.0000
0.0000  (   0.00%)
clock_network      0.6249            0.0000            0.0000
0.6249  (  62.98%)  i
register           0.2192        1.6331e-02          868.9040
0.2364  (  23.82%)
sequential         0.0000            0.0000            0.0000
0.0000  (   0.00%)
combinational  8.1739e-02        4.8843e-02          447.8848
0.1310  (  13.20%)
--------------------------------------------------------------------------
--------------------------
Total              0.9258 mW     6.5174e-02 mW     1.3168e+03 nW
0.9923 mW
```

Area

```
Number of ports:                          18
Number of nets:                           56
Number of cells:                          46
Number of combinational cells:            38
Number of sequential cells:                8
Number of macros/black boxes:              0
Number of buf/inv:                        19
Number of references:                      6

Combinational area:                83.066097
Buf/Inv area:                      31.443099
Noncombinational area:             82.596802
Macro/Black Box area:               0.000000
Net Interconnect area:      undefined  (No wire load specified)
```

```
Total cell area:                    165.662899
Total area:                  undefined
```

LFSR 12

14nm

Timing

```
  Point                              Trans       Incr        Path
  -----------------------------------------------------------------
  clock clk (rise edge)                          0.00        0.00
  clock network delay (ideal)                    0.00        0.00
  data_reg_0_/clocked_on (**FFGEN**)
                                      0.00       0.00        0.00 r
  data_reg_0_/Q (**FFGEN**)           0.00       0.00        0.00 r
  data[0] (out)                       0.00       0.00        0.00 r
  data arrival time                                          0.00

  clock clk (rise edge)                          0.10        0.10
  clock network delay (ideal)                    0.00        0.10
  output external delay                         -0.10        0.00
  data required time                                         0.00
  -----------------------------------------------------------------
  data required time                                         0.00
  data arrival time                                          0.00
  -----------------------------------------------------------------
  slack (MET)                                                0.00
```

Power

```
  Cell Internal Power  =   4.1737 uW   (58%)
  Net Switching Power  =   3.0521 uW   (42%)
                         ---------
Total Dynamic Power    =   7.2258 uW  (100%)

Cell Leakage Power     =   2.5555 nW


                 Internal         Switching           Leakage
Total
Power Group      Power            Power               Power
```

```
Power   (   %    )  Attrs
------------------------------------------------------------------------
--------------------------
io_pad              0.0000            0.0000            0.0000
0.0000  (   0.00%)
memory              0.0000            0.0000            0.0000
0.0000  (   0.00%)
black_box           0.0000            0.0000            0.0000
0.0000  (   0.00%)
clock_network       0.0000            0.0000            0.0000
0.0000  (   0.00%)  i
register            0.0000            1.8279            0.0000
1.8279  (  25.29%)
sequential          0.0000            0.0000            0.0000
0.0000  (   0.00%)
combinational       4.1737            1.2242        2.5555e+03
5.4004  (  74.71%)
------------------------------------------------------------------------
--------------------------
Total               4.1737 uW         3.0521 uW     2.5555e+03 pW
7.2284 uW
```

Area

```
Number of ports:                           26
Number of nets:                            53
Number of cells:                           39
Number of combinational cells:             27
Number of sequential cells:                12
Number of macros/black boxes:               0
Number of buf/inv:                          1
Number of references:                       6

Combinational area:                  8.302800
Buf/Inv area:                        0.177600
Noncombinational area:               0.000000
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (No wire load specified)

Total cell area:                     8.302800
Total area:                 undefined
```

32nm

Timing

```
Point                                 Trans      Incr        Path
 ------------------------------------------------------------------
clock clk (rise edge)                            0.00        0.00
clock network delay (ideal)                      0.00        0.00
data_reg_11_/CLK (DFFASRX1_RVT)
                                       0.00      0.00        0.00 r
data_reg_11_/Q (DFFASRX1_RVT)
                                       0.02      0.10        0.10 r
U45/SO (HADDX1_RVT)                    0.02      0.09        0.19 f
U46/Y (XOR3X2_RVT)                     0.03      0.06        0.25 r
data_reg_0_/D (DFFASRX1_RVT)
                                       0.03      0.01        0.26 r
data arrival time                                            0.26

clock clk (rise edge)                            0.30        0.30
clock network delay (ideal)                      0.00        0.30
data_reg_0_/CLK (DFFASRX1_RVT)                   0.00        0.30 r
library setup time                              -0.04        0.26
data required time                                           0.26
 ------------------------------------------------------------------
data required time                                           0.26
data arrival time                                           -0.26
 ------------------------------------------------------------------
slack (VIOLATED: increase significant digits)                0.00
```

Power

```
  Cell Internal Power  = 363.3564 uW   (94%)
  Net Switching Power  =  22.6895 uW    (6%)
                         ---------
Total Dynamic Power    = 386.0458 uW  (100%)

Cell Leakage Power     =  17.3839 uW


                 Internal         Switching           Leakage
Total
Power Group      Power            Power               Power
```

```
Power   (   %    )  Attrs
--------------------------------------------------------------------------
--------------------------
io_pad             0.0000            0.0000            0.0000
0.0000  (   0.00%)
memory             0.0000            0.0000            0.0000
0.0000  (   0.00%)
black_box          0.0000            0.0000            0.0000
0.0000  (   0.00%)
clock_network    303.0329            0.0000            0.0000
303.0329  (  75.11%)  i
register          26.3856            9.8685        1.2198e+07
48.4520  (  12.01%)
sequential         0.0000            0.0000            0.0000
0.0000  (   0.00%)
combinational     33.9378           12.8210        5.1859e+06
51.9447  (  12.88%)
--------------------------------------------------------------------------
--------------------------
Total            363.3564 uW        22.6894 uW     1.7384e+07 pW
403.4297 uW
```

Area

```
Number of ports:                           26
Number of nets:                            56
Number of cells:                           41
Number of combinational cells:             29
Number of sequential cells:                12
Number of macros/black boxes:               0
Number of buf/inv:                          3
Number of references:                       7

Combinational area:                 57.436545
Buf/Inv area:                        4.066304
Noncombinational area:              91.491840
Macro/Black Box area:                0.000000
Net Interconnect area:              11.965992

Total cell area:                   148.928385
Total area:                        160.894377
```

45nm

Timing

```
Point                              Trans     Incr      Path
------------------------------------------------------------
clock clk (rise edge)                        0.00      0.00
clock network delay (ideal)                  0.00      0.00
data_reg_3_/CLK (DFFSR)             0.00     0.00      0.00 r
data_reg_3_/Q (DFFSR)               0.04     0.13      0.13 r
U40/Y (XNOR2X1)                     0.05     0.06      0.19 r
U39/Y (XNOR2X1)                     0.02     0.04      0.22 f
U74/Y (INVX1)                       0.00     0.00      0.23 r
data_reg_0_/D (DFFSR)               0.00     0.00      0.23 r
data arrival time                                      0.23

clock clk (rise edge)                        0.30      0.30
clock network delay (ideal)                  0.00      0.30
data_reg_0_/CLK (DFFSR)                      0.00      0.30 r
library setup time                          -0.07      0.23
data required time                                     0.23
------------------------------------------------------------
data required time                                     0.23
data arrival time                                     -0.23
------------------------------------------------------------
slack (MET)                                            0.00
```

Power

```
  Cell Internal Power  =   1.3772 mW   (94%)
  Net Switching Power  =  84.5948 uW    (6%)
                         --------
Total Dynamic Power    =   1.4618 mW  (100%)

Cell Leakage Power     =   1.9546 uW


                 Internal         Switching           Leakage
Total
Power Group      Power            Power               Power
Power   (   %    )  Attrs
--------------------------------------------------------------------------
```

```
________________________________
io_pad               0.0000             0.0000             0.0000
0.0000  (    0.00%)
memory               0.0000             0.0000             0.0000
0.0000  (    0.00%)
black_box            0.0000             0.0000             0.0000
0.0000  (    0.00%)
clock_network        0.9373             0.0000             0.0000
0.9373  (   64.04%)  i
register             0.3309         2.0416e-02         1.3034e+03
0.3526  (   24.09%)
sequential           0.0000             0.0000             0.0000
0.0000  (    0.00%)
combinational        0.1090         6.4179e-02           651.2633
0.1738  (   11.88%)
________________________________________________________________________
________________________________
Total                1.3772 mW      8.4595e-02 mW      1.9546e+03 nW
1.4638 mW
```

Area

```
Number of ports:                                   26
Number of nets:                                    80
Number of cells:                                   66
Number of combinational cells:                     54
Number of sequential cells:                        12
Number of macros/black boxes:                       0
Number of buf/inv:                                 27
Number of references:                               6

Combinational area:                        116.855696
Buf/Inv area:                               46.460699
Noncombinational area:                     123.895203
Macro/Black Box area:                        0.000000
Net Interconnect area:      undefined  (No wire load specified)

Total cell area:                           240.750899
Total area:                 undefined
```